



Lab Sheet 1

Implementation of Discrete Time Signals using MATLAB

↖ Prerequisite

Before starting this section, students should know..

1. A fundamental knowledge on MATLAB
2. A theoretical knowledge on discrete time signals.

↗ Outcomes

After finishing this lab students should be able to.....

1. describe discrete-time signals mathematically and generate, manipulate, and plot discrete-time signals using MATLAB.
2. understand different operations on DT signals

➡ Outlines of this Lab

1. Lab Session 1.1: Generation of DT signals using MATLAB
2. Lab Session 1.2: Operation on DT signals
3. Lab Session 1.3: In Lab Evaluation
4. Home work

Lab Session 1.1:

LS 1.1.1 Generation of DT signals using MATLAB

Example 1.1

Generate the sequence

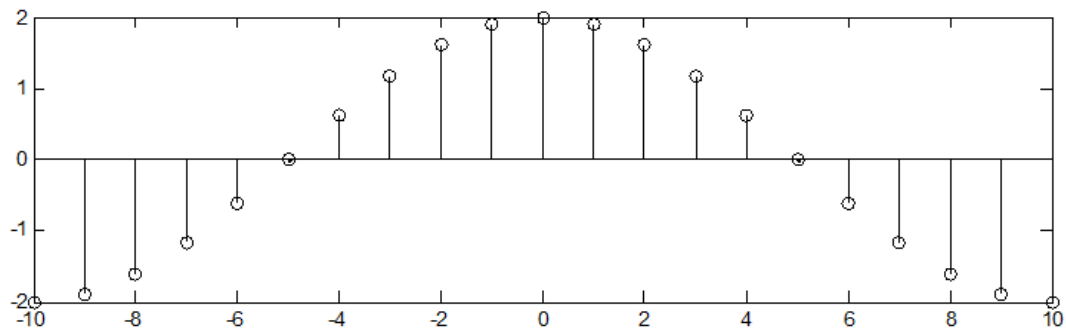
$$x(n) = 2\cos(2\pi 0.05n), -10 \leq n \leq 10$$

Solution:

MATLAB codes:

```
n=[-10:10];  
x=2*cos(2*pi*0.05*n);  
stem(n,x,'k')
```

Output:



Example 1.2

Generate a sequence $x(n) = [2 \ 1 \ -1 \ 3 \ 1 \ 4 \ 3 \ 7]$ represented in MATLAB.



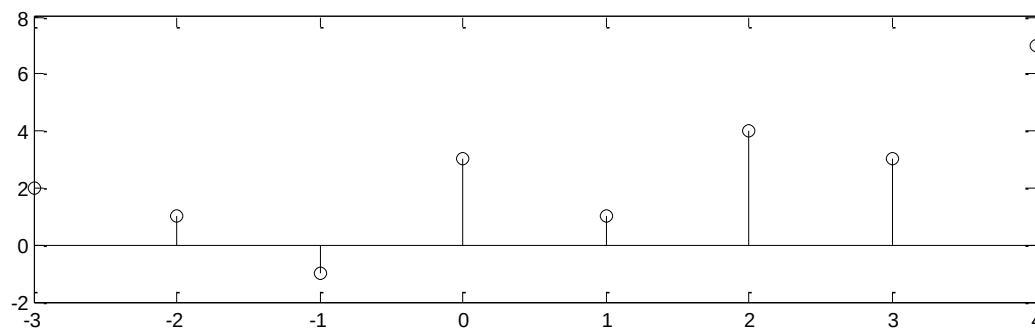
Solution:

MATLAB codes:

```
n=[-3 -2 -1 0 1 2 3 4];
x=[2 1 -1 3 1 4 3 7];
stem(n,x)
```

****Note the element $x(n=0)$**

Output:



Used custom function in this section:

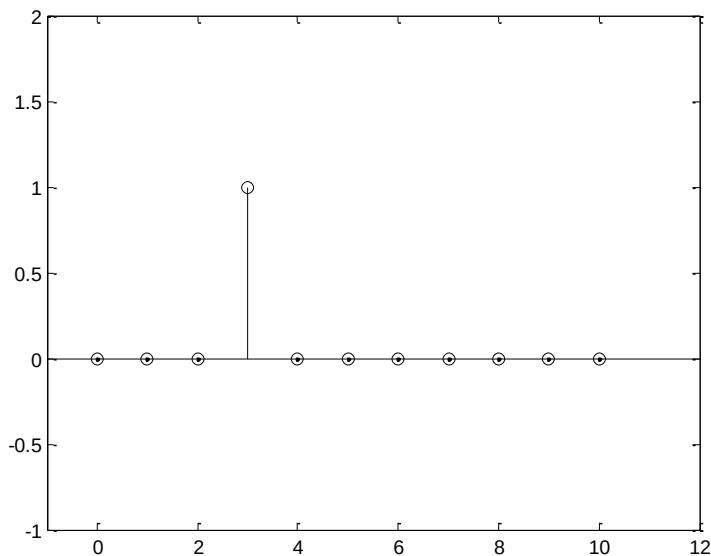
Table 1.1

	Name	Mathematical definition	MATLAB function
1.	Unit sample function	$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$	<code>function [x n]= delta(n0,n1,n2)</code> <code>n=[n1:n2];</code> <code>x=[n-n0]==0;</code>
2.	Unit step function	$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$	<code>function [x n]= u(n0,n1,n2)</code> <code>n=[n1:n2];</code> <code>x=[n-n0]>=0;</code>
3.	Real valued exponential sequence	$x(n) = a^n, \forall n; a \in \mathbb{R}$	<code>function [x n]= realx(a,n1,n2)</code> <code>n=[n1:n2];</code> <code>x=a.^n;</code>

Try the following codes

```
[x,n]=delta(3,0,10)
x =
    0    0    0    1    0    0    0    0    0    0    0
n =
    0    1    2    3    4    5    6    7    8    9   10

stem(n,x,'k'); axis([-1 12 -1 2])
```



Similarly try the following codes

```
a=2; n1=1; n2=10;
[x n]= realx(a,n1,n2)
stem(n,x,'k');
```

Example 1.3:

Generate and plot the following sequences over the indicated intervals.

(a) $x(n) = 2\delta(n+2) - \delta(n-4)$; $-5 \leq n \leq 5$.

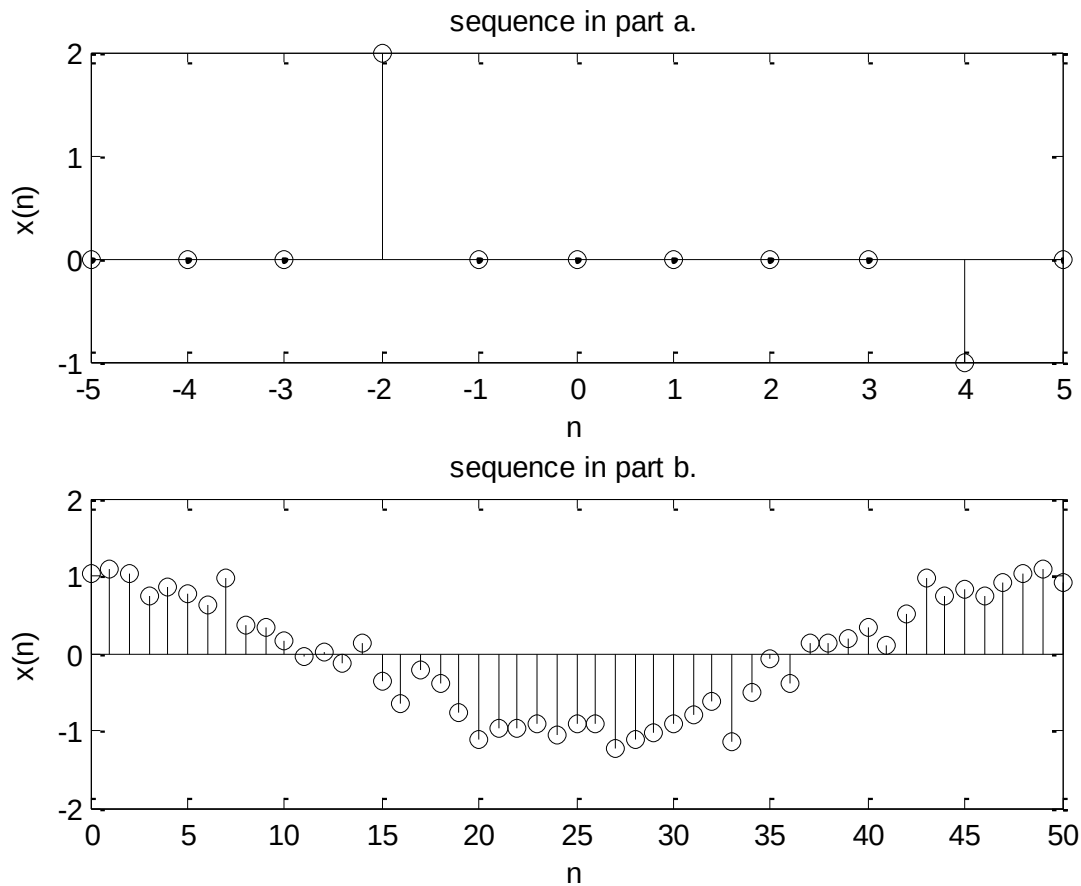
(b) $x(n) = \cos(0.04\pi n) + 0.2w(n)$, $0 \leq n \leq 50$, where $w(n)$ is the Gaussian random sequence with zero mean and unit variance.

Solution :

MATLAB Code:

```
%for part a
n=[-5:5];
x=2*delta(-2,-5,5)-delta(4,-5,5);
subplot(2,1,1);stem(n,x);
title('sequence in part a. ');xlabel('n');ylabel('x(n)');
%for part b
n=[0:50];
x=cos(0.04*pi*n)+0.2*randn(size(n));
subplot(2,1,2);stem(n,x);
title('sequence in part b. ');xlabel('n');ylabel('x(n)');
```

Output:



Example 1.4:

Generate and plot the samples (use the stem function) of the following sequence using MATLAB:

$$(a) x_1(n) = \sum_{m=0}^{10} (m+1) [\delta(n-2m) - \delta(n-2m-1)]; \quad 0 \leq n \leq 25$$

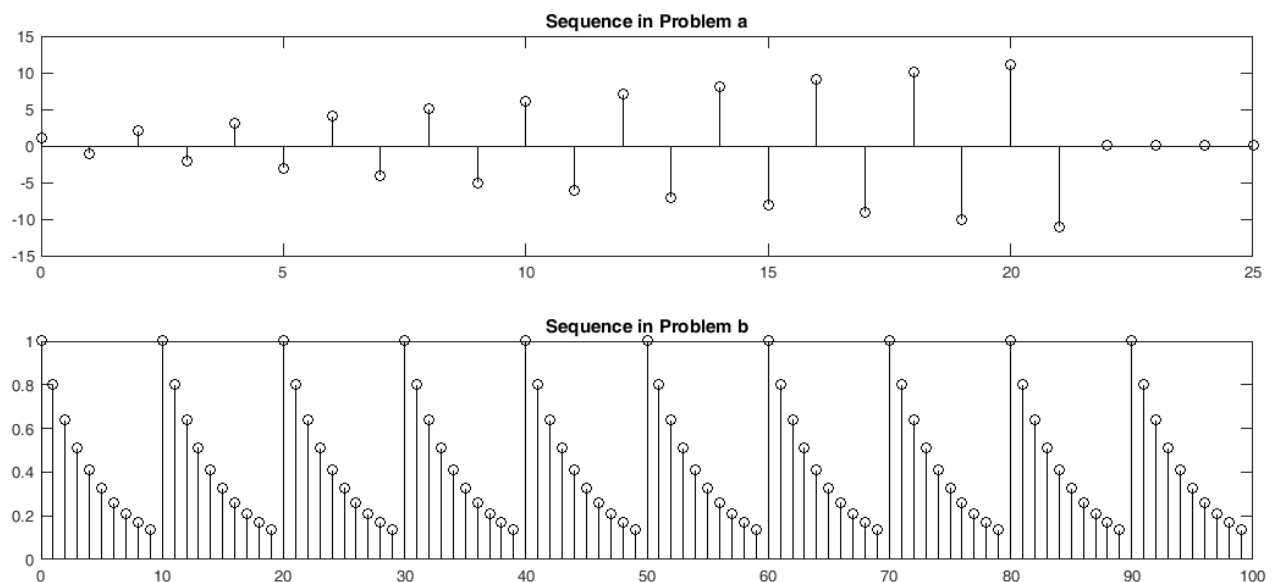
$$(b) x(n) = (0.8)^n, \quad 0 \leq n \leq 9, \text{ one periodic, plot 10 periods.}$$

Solution :

MATLAB Code:

```
% (a)
n = [0:25];
x1 = zeros(1,26);
for m=0:10,
    x1 = x1 + (m+1).*(delta(2.*m,0,25)-delta(2.*m+1,0,25));
end
subplot(2,1,1);
stem(n,x1,'k');
title('Sequence in Problem a')
% (b)
n=[0:9]';
x=(0.8).^n
N=10;
y = repmat(x,N,1); %repmat is a library function
m=0:1:size(y,1)-1;
subplot(2,1,2);
stem(m,y,'k')
title('Sequence in Problem b')
```

Output:



LS 1.2 Operation on DT Sequences

Used custom function in this section:

Table 1.2

	Name	Mathematical definition	MATLAB function
1.	Signal addition	$\{x_1(n)\} + \{x_2(n)\} = \{x_1(n) + x_2(n)\}$	<pre>function [y n]=sigadd(x1,n1,x2,n2) n=min(min(n1),min(n2)):max(max(n1),max(n2)); y1=zeros(1,length(n));y2=y1; y1(find((n>=min(n1)) & (n<=max(n1))==1))=x1; y2(find((n>=min(n2)) & (n<=max(n2))==1))=x2; y=y1+y2; end</pre>
2.	Signal multiplication	$\{x_1(n)\} \cdot \{x_2(n)\} = \{x_1(n) \cdot x_2(n)\}$	<pre>function [y n]=sigmult(x1,n1,x2,n2) n=min(min(n1),min(n2)):max(max(n1),max(n2)); y1=zeros(1,length(n));y2=y1; y1(find((n>=min(n1)) & (n<=max(n1))==1))=x1; y2(find((n>=min(n2)) & (n<=max(n2))==1))=x2; y=y1.*y2; end</pre>
3.	Signal Shifting	$y(n)=\{x(n-k)\}$	<pre>function [y n]=sigshift(x,m,n0) n=m+n0; y=x; end</pre>
4.	Signal folding	$y(n)=\{x(-n)\}$	<pre>function [y n] = sigfold(x,n) y=fliplr(x); n=-fliplr(n); % fliplr is library function. end >> help fliplr in command window to know in details .</pre>
5	Decomposition to even and odd part	$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$ $x_o(n) = \frac{1}{2}[x(n) - x(-n)]$	<pre>function [xe, xo, m] = evenodd(x,n) m = -fliplr(n); m1 = min([m,n]); m2 = max([m,n]); m = m1:m2; nm = n(1)-m(1); n1 = 1:length(n); x1 = zeros(1,length(m)); x1(n1+nm) = x; x = x1; xe = 0.5*(x + fliplr(x)); xo = 0.5*(x - fliplr(x)); end</pre>

Try the following codes

```
[x1,n1] = delta(3,0,10); subplot(3,1,1);stem(n1,x1);
[x2,n2] = delta(7,0,10); subplot(3,1,2);stem(n2,x2);
[x3,n3] = sigadd(x1*2,n1,-x2*3,n2);subplot(3,1,3);stem(n3,x3);
```

Example 1.5:

Let $x(n) = \{1, 2, 3, 4, 5, 6, 7, 6, 5, 4, 3, 2, 1\}$

Generate and plot the following sequence using MATLAB:

(a) $x_1(n) = 2x(n-5) - 3x(n+4)$

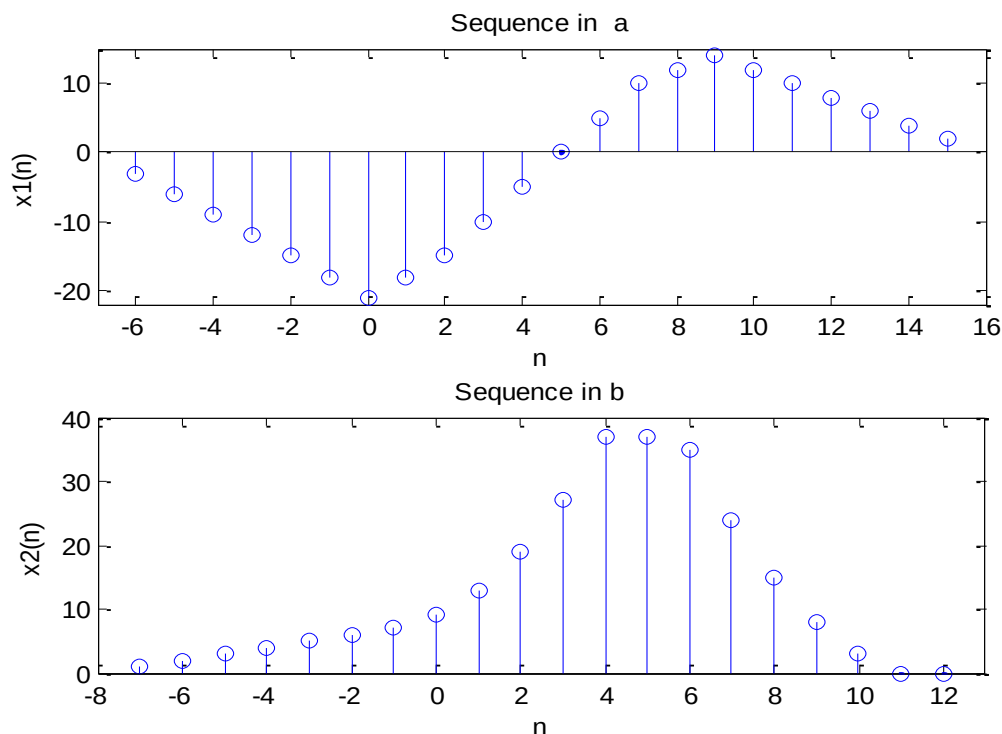
(b) $x_2(n) = x(3-n) + x(n)x(n-2)$

Solution :

MATLAB Code:

```
n = -2:10; x = [1:7,6:-1:1];
% a) x1(n) = 2*x(n-5) - 3*x(n+4)
[x11,n11] = sigshift(x,n,5); [x12,n12] = sigshift(x,n,-4);
[x1,n1] = sigadd(2*x11,n11,-3*x12,n12);
subplot(2,1,1);
stem(n1,x1); title('Sequence in a')
xlabel('n'); ylabel('x1(n)');
axis([min(n1)-1,max(n1)+1,min(x1)-1,max(x1)+1])
% b) x2(n) = x(3-n) + x(n)*x(n-2)
[x21,n21] = sigfold(x,n); [x21,n21] = sigshift(x21,n21,3);
[x22,n22] = sigshift(x,n,2); [x22,n22] = sigmult(x,n,x22,n22);
[x2,n2] = sigadd(x21,n21,x22,n22);
subplot(2,1,2);
stem(n2,x2);
title('Sequence in Example b')
xlabel('n');
ylabel('x2(n)');
axis([min(n2)-1,max(n2)+1,0,40])
```

Output :



Example 1.6: Real and Imaginary part of complex sequence

Generate the complex valued signal,

$$x(n) = e^{(-0.1+j0.3)n}, -10 \leq n \leq 10$$

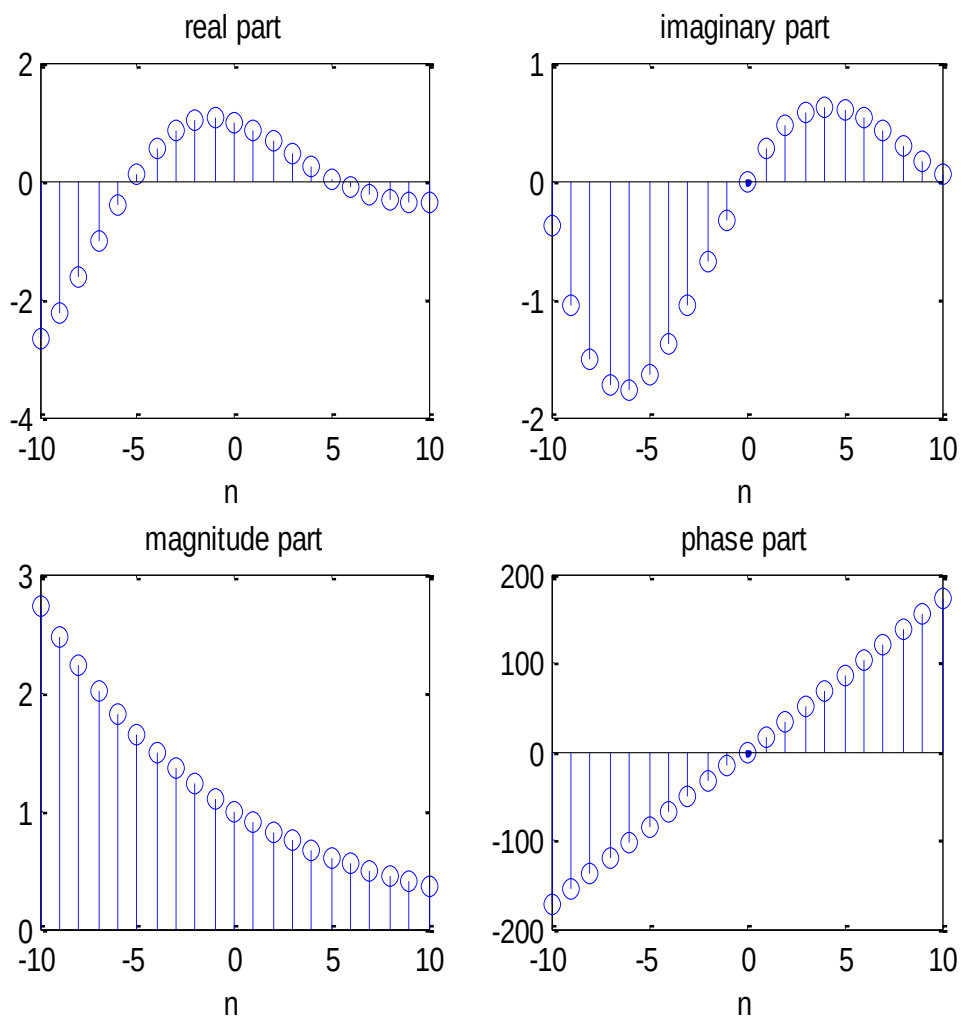
and plot its magnitude ,phase, the real part and the imaginary part in four separate subplots .

Solution:

MATLAB Code:

```
n = [-10:1:10]; alpha = -0.1+0.3j;  
x = exp(alpha*n);  
subplot(2,2,1);stem(n,real(x));title('real part');xlabel('n')  
subplot(2,2,2);stem(n,imag(x));title('imaginary part');xlabel('n')  
subplot(2,2,3);stem(n,abs(x));title('magnitude part');xlabel('n')  
subplot(2,2,4);stem(n,(180/pi)*angle(x));  
title('phase part');xlabel('n')
```

Output :



Example 1.7: Decomposing a sequence in even and odd components:

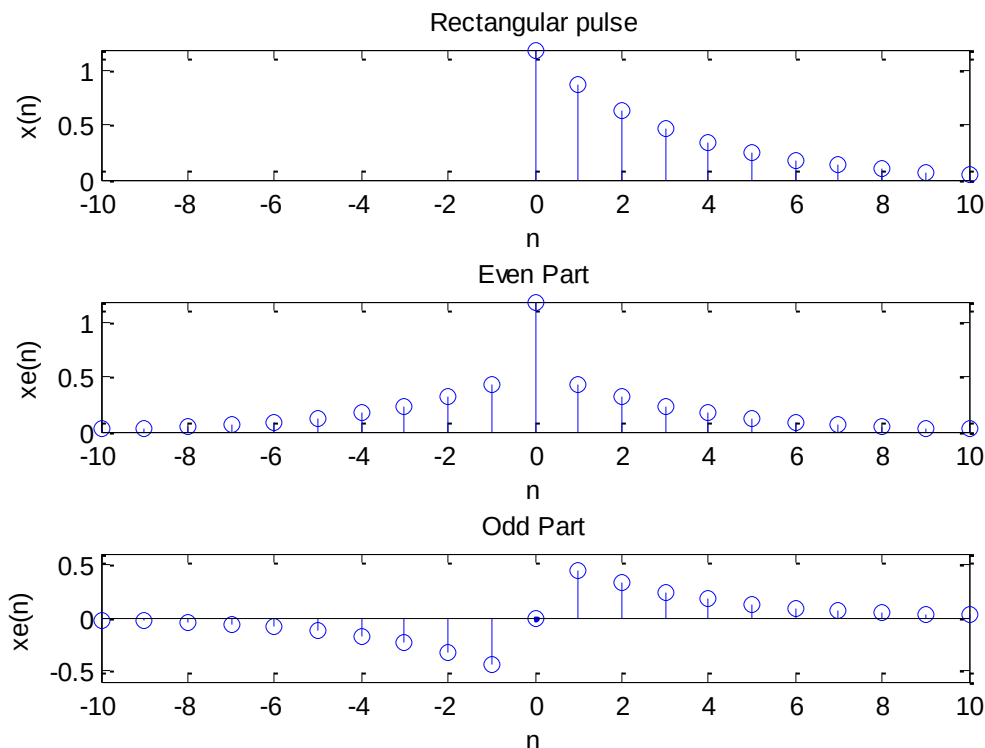
Use the function `evenodd(x,n)` to decompose $x(n) = 1.2e^{-0.1\pi n}$ into even and odd components. Show original sequence, even part and odd part in subplots.

Solution:

MATLAB Code:

```
n = [0:10];  
x=1.2*exp(-0.1*pi*n);  
[xe,xo,m] = evenodd(x,n);  
subplot(3,1,1); stem(n,x); title('Original Signal')  
xlabel('n'); ylabel('x(n)'); axis([-10,10,0,1.2])  
subplot(312); stem(m,xe); title('Even Part')  
xlabel('n'); ylabel('xe(n)'); axis([-10,10,0,1.2])  
subplot(313); stem(m,xo); title('Odd Part')  
xlabel('n'); ylabel('xe(n)'); axis([-10,10,-0.6,0.6])
```

Output:



Downsampling and Upsampling.

Downsampling:

Used library function:

```
downsample(x,k)
```

Example 1.8:

Use `downsample(x,k)` function to plot the original and downsampled sequence in subplots, where,

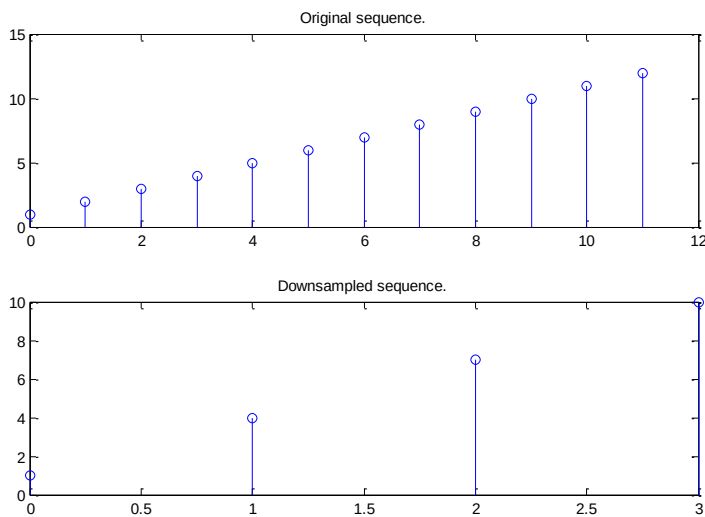
$x(n) = \{1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10 \ 11 \ 12\}$
and the downsampling factor is 3.

Solution:

MATLAB Code:

```
x = [ 1 2 3 4 5 6 7 8 9 10 11 12];  
k = 3;  
z=downsample(x,k)  
n=[0:length(x)-1];  
subplot(2,1,1)  
stem(n,x)  
title('Original sequence.')  
m=[0:length(z)-1];  
subplot(2,1,2)  
stem(m,z)  
title('Downsampled sequence.')
```

Output:



Upsampling:

Used library function:

upsample(x,k)

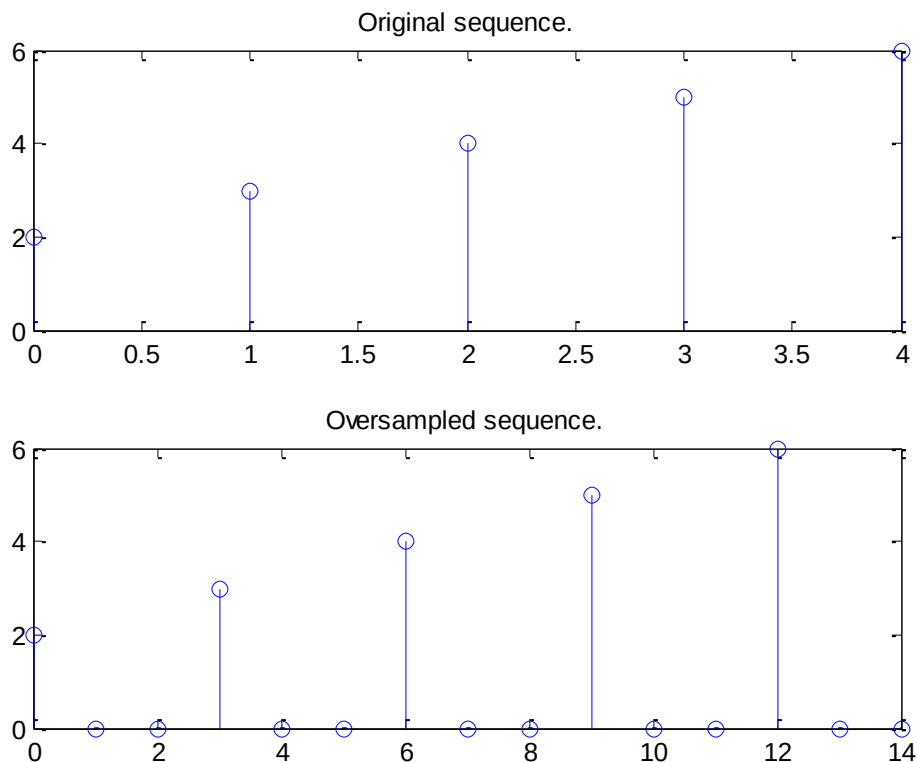
Example 1.9:

Use `upsample(x,k)` function to plot the original and downsampled sequence in subplots, where, $x(n) = \{2 \ 3 \ 4 \ 5 \ 6\}$ and the upsampling factor is 3.

Solution: MATLAB Code:

```
x = [ 2 3 4 5 6];  
k = 3;  
z=upsample(x,k);  
n=[0:length(x)-1];  
subplot(2,1,1)  
stem(n,x)  
title('Original sequence.')  
m=[0:length(z)-1];  
subplot(2,1,2)  
stem(m,z)  
title('Oversampled sequence.')  
stem(m,z)  
title('Original sequence.')
```

Output:



Lab Session 1.3 : In Lab Evaluation

In this section, students will solve the following problems in lab hour themselves.

ILE 1.1 : Using the `evenodd` function, decompose the following sequences into their even and odd components. Plot these components using the `stem` function.

$$x(n) = e^{0.1n} [u(n+5) - u(n-10)]$$

ILE 1.2 : Let $x(n) = \{1, -2, 4, 6, -5, 8, 10\}$. Generate and plot the samples of the following sequence:

$$y(n) = x(n+4)x(n-1) + 4x(n+4) + 3x(n)$$

ILE 1.3 : Let $x(n) = \{1, 2, 3, 4\}$ is one period of a periodic sequence. Plot 4 periods of the sequence.

Home Work:

1. Generate and plot the samples (use the `stem` function) of the following sequence using MATLAB : $x_1(n) = 3x(n+2) + x(n-4) - 2x(n)$ where, $x(n) = \{1, -2, 6, -5, 4, 6, 8, 10\}$
Hint: Use “`sigadd`” & “`sigshift`” functions. 

2. (a) A unit ramp DT signal $r(n)$ is defined as,

$$r(n) = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Develop a function to implement the above signal.

- (b) Use the above sequence to plot a ramp sequence for the interval -10 to 10.
3. Generate $x(n) = \cos(0.25\pi n)$, $-50 \leq n \leq 50$. Downsample $x(n)$ by a factor of 5 to generate $y(n)$. Plot both $x(n)$ and $y(n)$ using subplot.
4. Let, $x(n) = \{1, -2, 6, -5, 4, 6, 8, 10\}$ Generate and plot the samples of the following sequence:

$$y(n) = \sum_{k=1}^5 nx(n-k)$$

Hint: Use a for loop and “`sigadd`” & “`sigshift`” functions.

5. A simple digital differentiator is given by

$$y(n) = x(n) - x(n-1)$$

which computes a backward first-order difference of the input sequence. Implement this differentiator on the following sequences, and plot the results. Comment on the appropriateness of this simple differentiator.

(i) $x(n) = 5[u(n) - u(n-20)]$

(ii) $x(n) = \sin\left(\frac{\pi n}{25}\right)[u(n) - u(n-100)]$